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Type II cells. Type I cells are very thin so that gases can diffuse rapidly through them. Type II cells are thicker, synthesize and secrete a substance called **Surfactant**.

The lungs are light spongy tissues enclosed in the thoracic cavity surrounded by an airtight space. The thoracic cavity is bound dorsally by the vertebral column and ventrally by the sternum, laterally by the ribs and on the lower side by the dome shaped diaphragm.

The lungs are covered by double walled pleural membrane containing a several layers of elastic connective tissues and capillaries, which encloses the pleural fluid. Pleural fluid reduces friction when the lungs expand and contract.

Characteristic features of respiratory surface:

- surface area must be very large and richly supplied with blood vessels
- should be extremely thin and kept moist
- should be in direct contact with the environment
- should be permeable to respiratory gases

The steps involved in respiration are

i. The exchange of air between the atmosphere and the lungs.

ii. The exchange of O_2 and CO_2 between the lungs and the blood.

iii. Transport of O_2 and CO_2 by the blood.

iv. Exchange of gases between the blood and the cells.

v. Uptake of O_2 by the cells for various activities and the release of CO_2.

SURFACTANTS are the thin non–cellular films made of protein and phospholipids covering the alveolar membrane. The surfactant lowers the surface tension in the alveoli and prevents the lungs from collapsing. It also prevents pulmonary oedema. Premature Babies have low levels of surfactant in the alveoli may develop the new born respiratory distress syndrome (NRDS) because the synthesis of surfactants begins only after the 25th week of gestation.

6.3. Mechanism of breathing

The movement of air between the atmosphere and the lungs is known as ventilation or breathing. Inspiration and expiration are the two phases of breathing. Inspiration is the movement of atmospheric air into the lungs and expiration is the movement of alveolar air that diffuse out of the lungs. (Figure 6.4)

Lungs do not contain muscle fibres but expands and contracts by the movement of the ribs and diaphragm. The diaphragm is a sheet of tissue which separates the thorax from the abdomen. In a relaxed state, the diaphragm is domed shaped.

Observe a live fish and find out how many times it beats the operculum per minute. Now check your rate of breathing for a minute. The rate of breathing will be more in fish than you – Give reasons.